DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester January-April 2025)

Exploratory Data Analysis of District-wise IPC Crimes in India

Submitted by

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Course Code INT375

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DECLARATION

I, Nishant Raj, student of B. Tech CSE under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12/04/25 Signature

Registration No. 12308423 Nishant Raj

CERTIFICATE

This is to certify that Nishant Raj bearing Registration no. 12308423 has completed CSE375 project titled, **"Exploratory Data Analysis of District-wise IPC Crimes in India (2017–2022)"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

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I would also like to acknowledge the Government of India, particularly the National Crime Records Bureau (NCRB), for providing access to the district-wise IPC crime dataset. This publicly available dataset served as the foundation of this project and enabled a meaningful exploration of crime patterns, regional variations, and year-wise trends across different states and districts of India.

My sincere appreciation also goes to the open-source software community for developing powerful tools and libraries such as Python, Pandas, Matplotlib, and Seaborn. These technologies were essential for performing data cleaning, data transformation, visualization, and statistical analysis, which made it possible to derive meaningful insights from the dataset.

Finally, I would like to thank my peers, friends, and family members for their constant support and motivation. Their encouragement helped me remain focused and motivated throughout the project, especially during challenging stages of analysis and report preparation.

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1. INTRODUCTION

Crime analysis plays an essential role in understanding patterns of criminal activities and supporting effective law enforcement and policy formulation. With the increasing availability of publicly accessible datasets, data-driven approaches have become valuable tools for studying crime trends and identifying areas that require attention. This project focuses on the Exploratory Data Analysis of District-wise IPC Crimes in India from 2017 to 2022, using Python-based data analysis techniques to uncover meaningful insights from crime statistics.

The Indian Penal Code (IPC) forms the foundation of criminal law in India, covering a wide range of offences including violent crimes, property crimes, and offences against individuals. Analyzing district-wise IPC crime data provides a deeper understanding of how crime patterns vary across regions and over time. Such analysis helps identify high-crime districts, dominant crime categories, and year-wise changes in criminal activities.

Using tools such as Python, Pandas, Matplotlib, and Seaborn, this project performs systematic data cleaning, processing, and visualization of crime statistics obtained from the National Crime Records Bureau (NCRB) dataset. The study explores trends in major crimes, compares crime levels across states and districts, and examines relationships between different crime categories through correlation analysis.

The insights obtained from this analysis contribute to a better understanding of crime distribution across India and highlight the importance of data analytics in supporting evidence-based decision making for crime prevention, law enforcement planning, and public safety improvements.

2. SOURCE OF DATASET

https://indiadataportal.com/p/crime-statistics/r/ncrb-cii_ipc_crimes-dt-yr-aaa

3. EDA PROCESS

The Exploratory Data Analysis (EDA) process for this project involved several systematic steps to understand the structure of the dataset, ensure data quality, and identify meaningful patterns within the district-wise IPC crime statistics.

Data Understanding and Initial Exploration

The analysis began by loading the dataset using the Pandas library in Python. Initial exploration was performed to understand the structure and characteristics of the data.

- The dataset was inspected using functions such as. head(), .info(), .describe(), and .shape() to understand the number of records, columns, and data types.

- Important columns such as state_name, district_name, year, and various IPC crime categories were identified for further analysis.
- Basic descriptive statistics were examined to understand the distribution of numerical crime variables.

Data Cleaning

To ensure the reliability and accuracy of the analysis, several data cleaning steps were performed.

- The column rioting_vigilants was removed due to the presence of excessive missing values.
- Missing values in the atmpt_acid_attack column were handled using median imputation to maintain numerical consistency.
- Duplicate records were removed to avoid redundant observations.
- Rows containing negative values in numerical crime-related columns were filtered out to ensure logical data integrity.
- Data types were verified and numerical formats were standardized for accurate computation.

Numerical Validation

After cleaning the dataset, additional checks were performed to confirm the integrity of the data.

- The dataset was examined to ensure that no negative values remained in numerical columns.
- Summary statistics and uniqueness checks were used to verify the consistency and correctness of the cleaned dataset.

Visualization and Initial Statistical Analysis

To better understand crime patterns and distributions, several visual and statistical analysis techniques were applied.

- Histograms were used to observe the distribution of specific crime categories.
- Bar plots were created to identify districts and states with the highest crime counts.
- Heatmaps were used to visualize correlations between different crime categories.
- Summary statistics and uniqueness checks were computed using functions such as .describe() and .nunique() to gain additional insights into the dataset.

These exploratory analysis steps provided a strong foundation for identifying crime trends, regional patterns, and relationships between different crime categories within the dataset.

4. ANALYSIS ON DATASET

Objective 1: Year-wise Trend Analysis of Major Crimes

i. Introduction

Analyzing crime trends over time is essential for understanding how criminal activities evolve and for identifying possible patterns or fluctuations. Major crimes such as murder, hit-and-run incidents, and arson are particularly important indicators of public safety and law enforcement effectiveness. Studying year-wise trends of these crimes helps identify whether crime levels are increasing, decreasing, or remaining stable over time.

ii. General Description

In this analysis, the dataset was grouped by year, and the total number of selected crimes—murder, hit-and-run, and arson—was calculated for each year from 2017 to 2022. This allowed for the observation of trends and variations in these crime categories over the six-year period.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed to conduct the analysis:

- The dataset was grouped by the year column using the .groupby('year') function.
- The total counts for selected crime categories (murder, hit_and_run, arson) were calculated using .sum().
- A line plot was created using the Seaborn library (sns.lineplot) to visualize changes in crime levels over time.

iv. Analysis Results

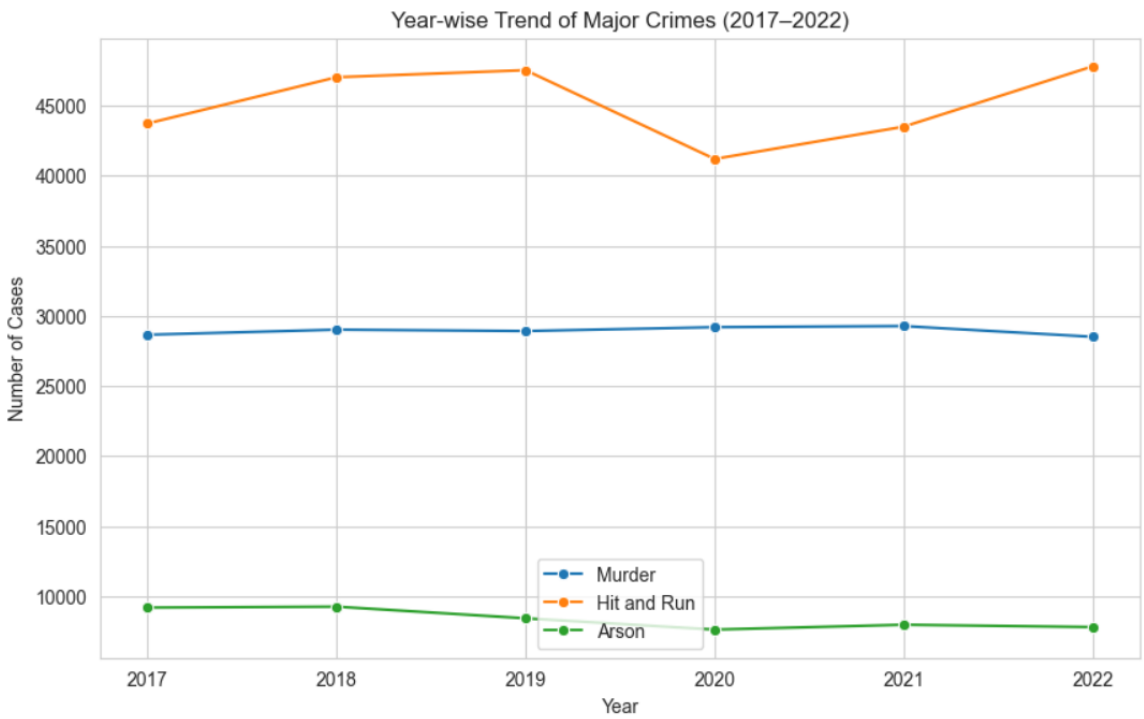
The results of the year-wise aggregation showed the following trends:

- The number of murder cases remained relatively stable, fluctuating slightly between approximately 28,000 and 29,000 cases per year.
- Hit-and-run incidents showed noticeable variation, with a decline around 2020, followed by an increase in subsequent years.
- Arson cases showed a gradual decline after 2018, suggesting a slight reduction in this category over the observed period.

Overall, the analysis indicates that while certain crime categories experience yearly fluctuations, the overall trend for major violent crimes remains relatively stable across the six-year period.

v. Visualization

The following visualization illustrates the year-wise trend of major crimes using a line plot.



Objective 2: Top 10 Districts and States with Highest Total IPC Crimes

i. Introduction

Identifying regions with high crime rates is essential for understanding geographic crime distribution and for assisting law enforcement agencies in prioritizing resources. Analyzing district-level and state-level crime totals helps highlight areas where crime incidents are more concentrated.

ii. General Description

In this step, the total number of IPC crimes was calculated for each district and each state by summing all crime-related columns in the dataset. The districts and states were then ranked based on their total crime counts, and the top 10 districts and top 10 states with the highest crime totals were identified.

iii. Specific Requirements, Functions, and Formulas

The following operations were performed:

- A new feature total_crimes was created by summing all crime-related columns in each row.

- District-level crime totals were calculated using .groupby('district_name').sum().
- State-level crime totals were calculated using .groupby('state_name').sum().
- The top 10 districts and states were identified using .sort_values(ascending=False).head(10).
- The results were visualized using bar plots.

iv. Analysis Results

The analysis revealed that several large metropolitan districts recorded the highest total number of crimes. Districts such as Mumbai, North West (Delhi), Pune, Hyderabad, Chennai, and Bengaluru Urban were among the top districts with the highest IPC crime totals.

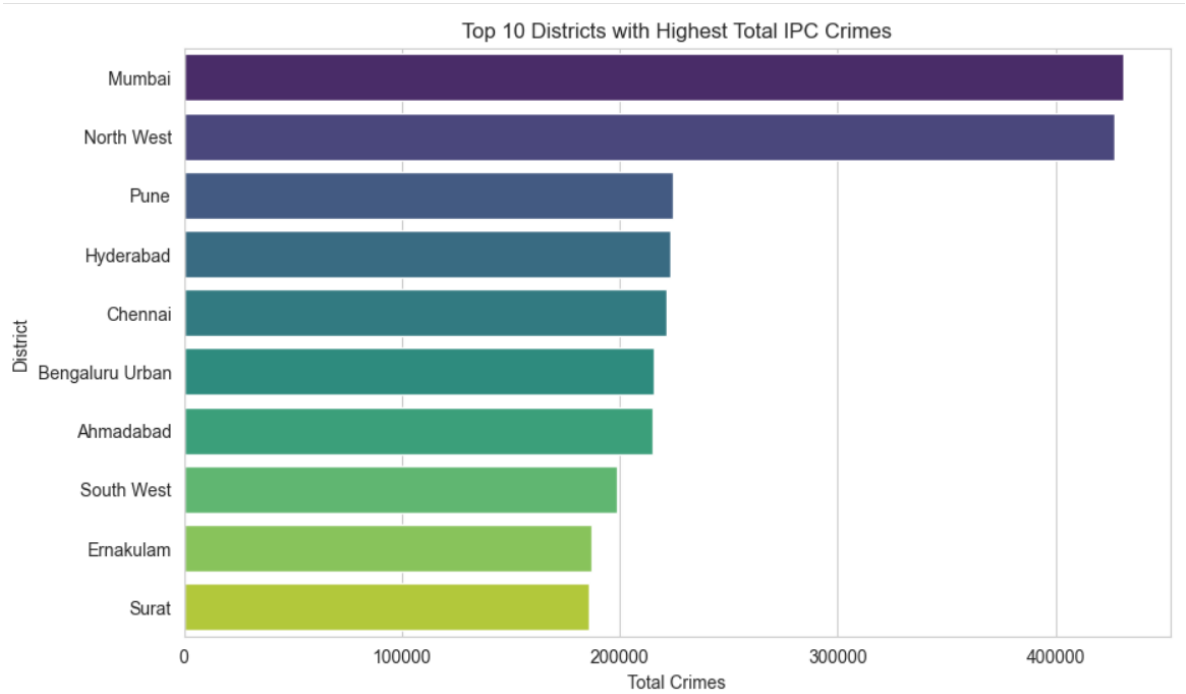
At the state level, the highest crime totals were observed in Uttar Pradesh, Maharashtra, and Tamil Nadu, followed by Madhya Pradesh, Delhi, and Rajasthan. These states have large populations and major urban centers, which often contribute to higher recorded crime counts.

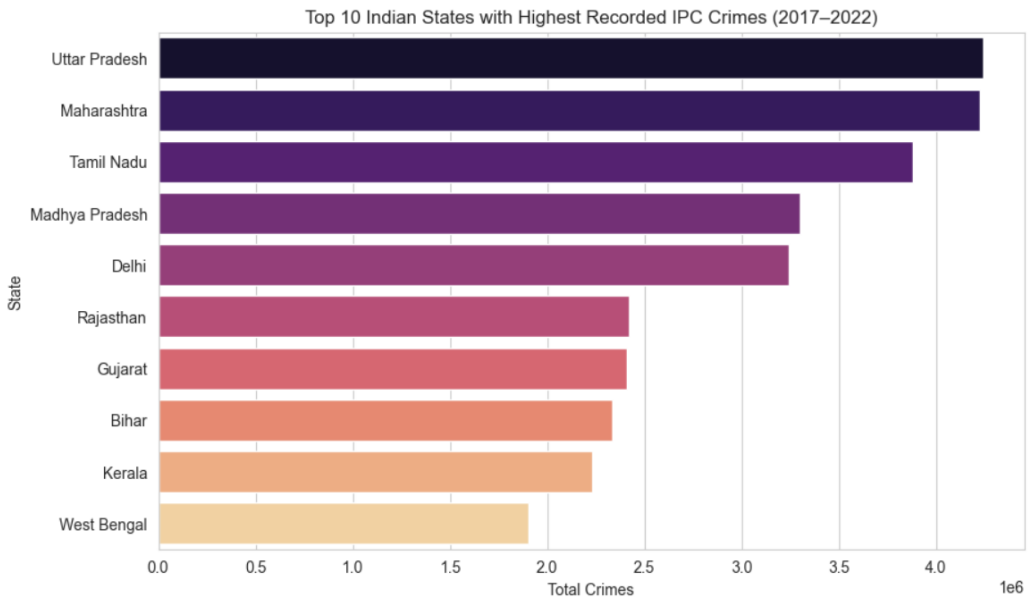
The findings highlight that crime distribution is not uniform across regions and tends to be higher in densely populated urban districts and major metropolitan areas.

v. Visualization

The following bar charts display:

• The top 10 districts with the highest total IPC crimes
• The top 10 states with the highest total IPC crimes





Objective 3: State-wise Crime Distribution Heatmap

i. Introduction

Visualizing crime distribution across states helps identify regions with higher crime burdens and highlights geographical disparities in crime patterns. A heatmap provides an effective way to compare crime totals across different states and quickly identify areas with higher concentrations of criminal activity.

ii. General Description

In this analysis, the total number of IPC crimes was aggregated for each state by summing the total_crimes variable across all districts and years. The resulting state-wise totals were then visualized using a heatmap to illustrate the relative crime levels across different states in India.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed:

- The dataset was grouped by state_name using the .groupby('state_name') function.
- The total number of crimes for each state was calculated using .sum().
- The states were sorted in descending order based on their total crime counts.
- A heatmap visualization was created using Seaborn (sns.heatmap) to represent the intensity of crime levels across states.

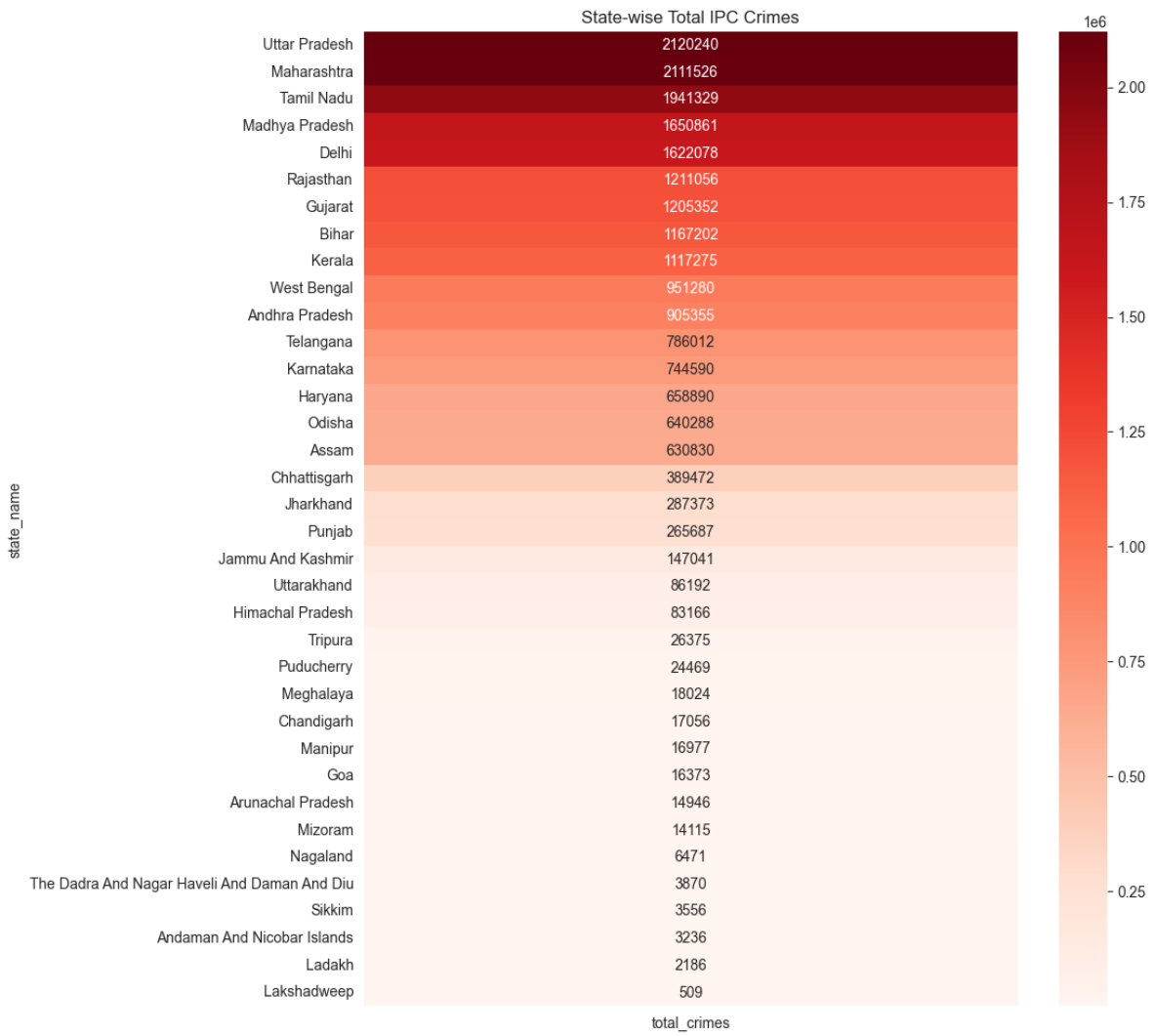
iv. Analysis Results

The heatmap revealed significant variation in crime distribution across Indian states. States such as Uttar Pradesh, Maharashtra, and Tamil Nadu recorded the highest total IPC crime counts during the period of analysis. Other states including Madhya Pradesh, Delhi, and Rajasthan also showed relatively high crime totals.

These findings suggest that crime reporting is generally higher in densely populated states and states containing major urban centers. The visualization highlights clear regional differences in crime patterns, which can assist policymakers and law enforcement agencies in identifying high-priority regions for crime prevention strategies.

v. Visualization

The heatmap below represents the state-wise distribution of total IPC crimes, where darker shades indicate higher crime totals.



Objective 4: Crime Type Distribution in a Selected State

i. Introduction

Understanding the distribution of different crime types within a particular state provides deeper insight into the nature of criminal activities occurring in that region. Identifying the most frequent crime categories can help authorities design targeted policies and crime prevention strategies.

ii. General Description

For this analysis, the dataset was filtered to include only records from the state of West Bengal. The total counts of all crime categories were then aggregated, and the top 10 crime categories with the highest number of cases were identified.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed:

- The dataset was filtered using the condition state_name == ' Andhra Pradesh.
- Crime counts across numerical crime-related columns were summed using .sum().
- The crime categories were sorted in descending order to identify the top 10 most common crime types.
- A bar chart visualization was created using Seaborn (sns.barplot) to clearly represent the most frequent crimes.

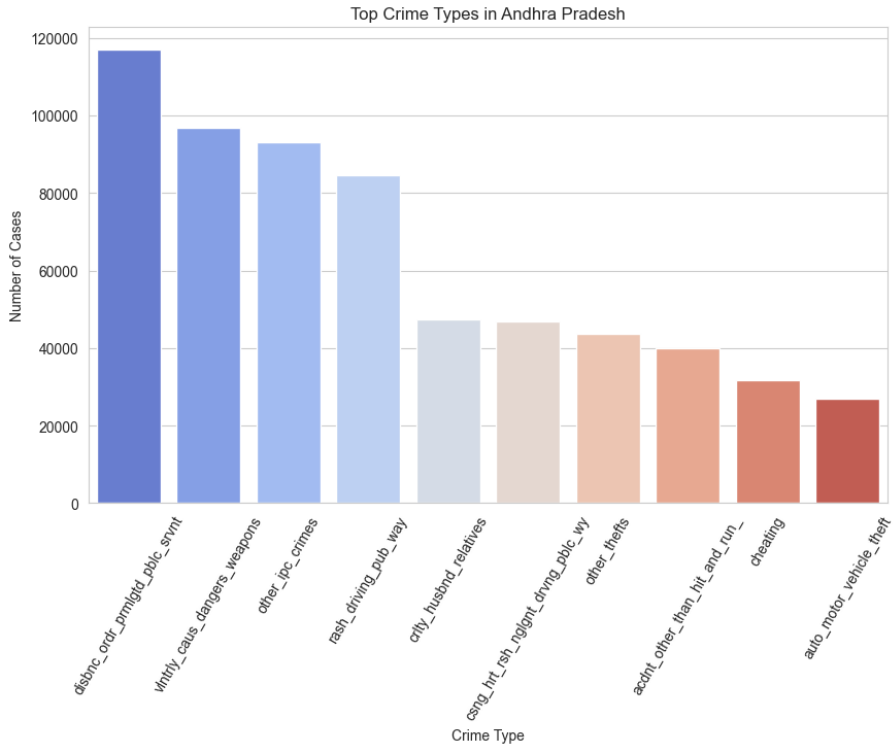
iv. Analysis Results

The analysis revealed that certain crime categories dominate the crime distribution within West Bengal. Categories such as other IPC crimes, theft-related offences, assault-related crimes, and criminal intimidation appear among the most frequently reported crimes in the state.

These results suggest that property crimes and interpersonal conflicts form a significant portion of the recorded criminal activities. Understanding these dominant crime categories can help authorities prioritize crime prevention strategies and allocate law enforcement resources more effectively.

v. Visualization

The bar chart below illustrates the top 10 crime categories reported in Andhra Pradesh, providing a clear representation of the most common offences recorded in the dataset.



Objective 5: Correlation Between Different Crime Categories

i. Introduction

Analyzing correlations between different crime categories helps identify potential relationships and patterns among various criminal activities. Such analysis can reveal whether certain crimes tend to occur together, which may indicate common underlying causes or similar social and economic factors.

ii. General Description

In this analysis, selected crime categories were examined to determine how strongly they are related to one another. Pearson correlation coefficients were calculated for the chosen crime variables, and the results were visualized using a heatmap to easily interpret the relationships between different crimes.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed:

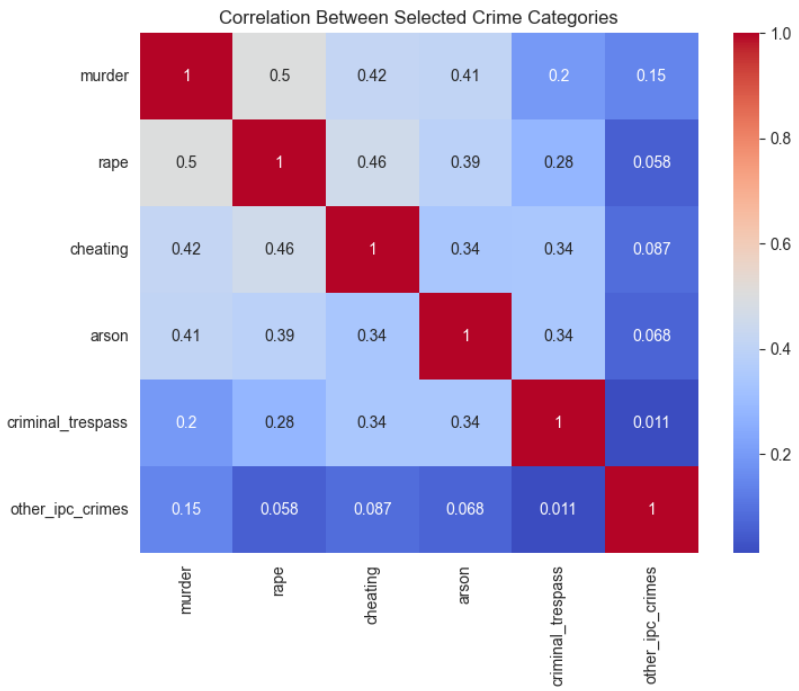
- Selected a subset of relevant crime categories including murder, hit-and-run, arson, cheating/impersonation, and criminal trespass.
- Calculated the Pearson correlation matrix using the .corr() function.
- Visualized the correlation values using a Seaborn heatmap (sns.heatmap) to clearly display relationships between crime variables.

iv. Analysis Results

The correlation analysis revealed moderate relationships among some crime categories, suggesting that certain crimes may occur together or share similar contributing factors. However, the correlations were not extremely strong, indicating that many crime types occur independently and are influenced by different circumstances.

v. Visualization

The heatmap below illustrates the correlation matrix between selected crime categories, where darker colors indicate stronger positive relationships.



Objective 6: Crime Distribution - Murder Cases

i. Introduction

Studying the distribution of specific crimes across districts helps understand how crime intensity varies geographically. In particular, analyzing the distribution of murder cases provides insight into how frequently this serious violent crime occurs across different regions.

ii. General Description

In this step, the distribution of murder cases across all districts and years was examined. A histogram was used to visualize how murder cases are distributed across districts and to identify whether most districts experience similar levels of crime or whether extreme cases exist.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed:

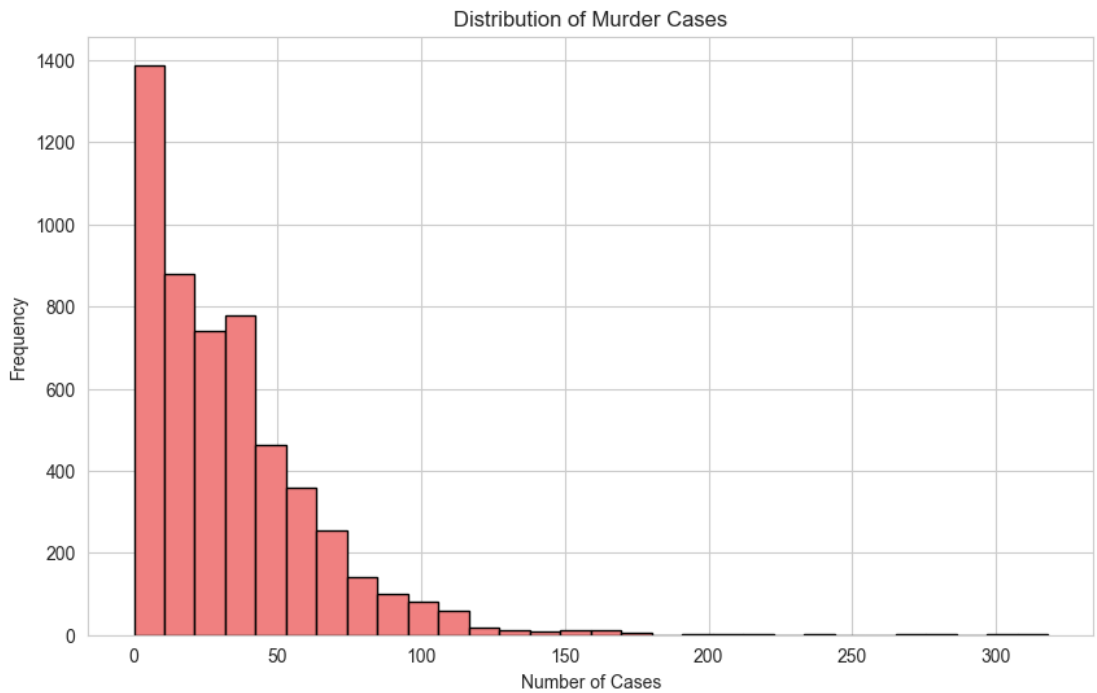
- Selected the murder column from the dataset.
- Plotted a histogram using the .hist() function to observe the distribution pattern.
- Used multiple bins to better visualize the spread of murder cases across districts.

iv. Analysis Results

The distribution of murder cases showed that most districts report relatively low numbers of murder incidents, while a small number of districts report significantly higher values. These higher values act as outliers, suggesting that certain districts experience disproportionately high levels of violent crime compared to others.

v. Visualization

The histogram below illustrates the distribution of murder cases across districts.



Objective 7: Crime Category Distribution

i. Introduction

Understanding the composition of different crime categories provides insight into which types of offences contribute the most to overall crime statistics. A category-wise analysis helps highlight the relative proportion of various crimes recorded under the Indian Penal Code.

ii. General Description

In this analysis, selected crime categories were aggregated to determine their overall contribution to the total crime count. The results were visualized using a pie chart, which helps illustrate the relative share of each crime category.

iii. Specific Requirements, Functions, and Formulas

The following steps were performed:

- Selected key IPC crime categories such as murder, arson, hit-and-run, criminal intimidation, insult to modesty of women, cruelty by husband or relatives, and other IPC crimes.
- Calculated the total number of cases for each category using the .sum() function.
- Visualized the distribution using a pie chart (plt.pie) to represent the proportional contribution of each crime category.

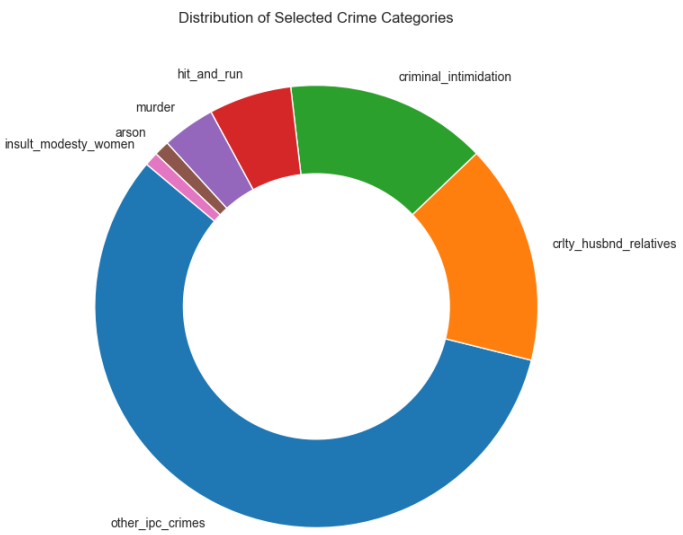
iv. Analysis Results

The results indicate that categories such as other IPC crimes and cruelty by husband or relatives account for a substantial proportion of the total crime cases. This highlights the significant role of domestic disputes, interpersonal conflicts, and miscellaneous IPC offences in the overall crime composition.

The analysis demonstrates that crime patterns are influenced not only by violent crimes but also by social and domestic issues, which form a large share of recorded criminal cases.

v. Visualization

The pie chart below illustrates the distribution of selected crime categories within the dataset.



5. CODING PART

```python
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```python
# Load Dataset
df = pd.read_csv(r"C:\Projects\Data Science\ipc-crime-data-analysis\data\districtwise-ipc-crimes-2017-onwards.csv")

print("Dataset Shape:", df.shape)
df.head()
```

```python
# Dataset Overview

print("\nDataset Info:\n")
print(df.info())

print("\nDataset Description:\n")
print(df.describe().round(2))

print("\nColumns:\n", df.columns.tolist())
```

```python
# Checking Missing Values

missing_values = df.isnull().sum().sort_values(ascending=False)

print("\nMissing Values per Column:")
print(missing_values[missing_values > 0])

print("\nTotal Missing Values:", df.isnull().sum().sum())
```

```python
# Handling Missing Values

df = df.drop(['rioting_vigilants'], axis=1)

df['atmpt_acid_attack'] = df['atmpt_acid_attack'].fillna(
    df['atmpt_acid_attack'].median()
)

# Remove duplicate rows
df = df.drop_duplicates()

# Remove rows with negative values
num_cols = df.select_dtypes(include=[np.number]).columns
df = df[(df[num_cols] >= 0).all(axis=1)]

print("Cleaned dataset shape:", df.shape)
```

```python
# Feature Engineering: Total Crimes

crime_columns = df.select_dtypes(include=np.number).columns
crime_columns = crime_columns.drop(['id','year','state_code','district_code'])

df['total_crimes'] = df[crime_columns].sum(axis=1)

df[['state_name','district_name','year','total_crimes']].head()
```

```python
# Year-wise Trend of Major Crimes

yearly_trends = df.groupby('year')[['murder','hit_and_run','arson']].sum().reset_index()

plt.figure(figsize=(10,6))

sns.lineplot(data=yearly_trends, x="year", y="murder", marker='o', label='Murder')
sns.lineplot(data=yearly_trends, x="year", y="hit_and_run", marker='o', label='Hit and Run')
sns.lineplot(data=yearly_trends, x="year", y="arson", marker='o', label='Arson')

plt.title("Year-wise Trend of Major Crimes (2017-2022)")
plt.xlabel("Year")
plt.ylabel("Number of Cases")
plt.legend()
plt.show()
```

```python
# Top 10 Districts with Highest Crimes

top_districts = (
    df.groupby('district_name')['total_crimes']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10,6))

sns.barplot(
    x=top_districts.values,
    y=top_districts.index,
    hue=top_districts.index,
    palette="viridis",
    legend=False
)

plt.title("Top 10 Districts with Highest Total IPC Crimes")
plt.xlabel("Total Crimes")
plt.ylabel("District")
plt.show()
```

```python
# Top 10 States with Highest Crimes

top_states = (
    df.groupby('state_name')['total_crimes']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(10,6))

sns.barplot(
    x=top_states.values,
    y=top_states.index,
    hue=top_states.index,
    palette="magma",
    legend=False
)
plt.title("Top 10 Indian States with Highest Recorded IPC Crimes (2017-2022)")
plt.xlabel("Total Crimes")
plt.ylabel("State")
plt.show()
```

```python
# State-wise Crime Heatmap

state_crimes = (
    df.groupby('state_name')['total_crimes']
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10,12))

sns.heatmap(
    state_crimes.to_frame(),
    annot=True,
    fmt=".0f",
    cmap="Reds"
)

plt.title("State-wise Total IPC Crimes")
plt.show()
```

```python
# Crime Distribution in a Selected State

state = "Andhra Pradesh"

state_data = df[df['state_name'] == state]

crime_distribution = (
    state_data[crime_columns]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

crime_df = crime_distribution.reset_index()
crime_df.columns = ['Crime_Type','Cases']

plt.figure(figsize=(10,6))

sns.barplot(
    x='Crime_Type',
    y='Cases',
    hue='Crime_Type',
    data=crime_df,
    palette='coolwarm',
    legend=False
)

plt.title(f"Top Crime Types in {state}")
plt.xticks(rotation=60)
plt.ylabel("Number of Cases")
plt.xlabel("Crime Type")

plt.show()
```

```python
# Correlation Between Crime Categories

selected_columns = [
    'murder',
    'rape',
    'cheating',
    'arson',
    'criminal_trespass',
    'other_ipc_crimes'
]

correlation_matrix = df[selected_columns].corr()

plt.figure(figsize=(8,6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Between Selected Crime Categories")

plt.show()
```

```python
# Distribution of Murder Cases

plt.figure(figsize=(10,6))

df['murder'].hist(
    bins=30,
    color='lightcoral',
    edgecolor='black'
)

plt.title('Distribution of Murder Cases')
plt.xlabel('Number of Cases')
plt.ylabel('Frequency')

plt.show()
```

```python
# Crime Category Distribution

crime_types = [
    'murder',
    'arson',
    'hit_and_run',
    'criminal_intimidation',
    'insult_modesty_women',
    'crlty_husbnd_relatives',
    'other_ipc_crimes'
]

category_totals = df[crime_types].sum().sort_values(ascending=False)

plt.figure(figsize=(8,8))

plt.pie(
    category_totals,
    labels=category_totals.index,
    startangle=140,
    wedgeprops=dict(width=0.4)
)

plt.title('Distribution of Selected Crime Categories')

plt.show()
```

6. OUTPUT AND VISUALIZATION

	id	year	state_name	state_code	district_name	district_code	registration_circles	murder	clpbl_hmcrd_not_amt_murder	hit_and_run	...	ofnc_religion	cheating_impersonation	ofnc_mischi
0	0	2017	Andhra Pradesh	28	Ananthapuramu	502	Anantapur	113.0	4.0	50.0	...	6.0	4.0	9
1	1	2017	Andhra Pradesh	28	Chittoor	503	Chittoor	70.0	5.0	36.0	...	0.0	0.0	4
2	2	2017	Andhra Pradesh	28	Y.S.R.	504	Cuddapah	88.0	10.0	32.0	...	0.0	1.0	44
3	3	2017	Andhra Pradesh	28	East Godavari	505	East Godavari	69.0	14.0	138.0	...	12.0	0.0	25
4	4	2017	Andhra Pradesh	28	Ananthapuramu	502	Guntakal Railway	11.0	0.0	0.0	...	0.0	0.0	0

5 rows × 124 columns

```
Dataset Info:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5322 entries, 0 to 5321
Columns: 124 entries, id to other_ipc_crimes
dtypes: float64(117), int64(4), object(3)
memory usage: 5.0+ MB
None

Dataset Description:

             id     year  state_code  district_code   murder  \
count  5322.00  5322.00     5322.00        5322.00  5322.00
mean   2660.50  2019.56       17.72         367.00    32.61
std    1536.47     1.71        9.89         386.28    30.21
min       0.00  2017.00        1.00           1.00     0.00
25%    1330.25  2018.00        9.00         171.00    10.00
50%    2660.50  2020.00       19.00         356.00    27.00
75%    3990.75  2021.00       24.00         536.75    46.00
max    5321.00  2022.00       38.00        9999.00   318.00

       clpbl_hmcrd_not_amt_murder  hit_and_run  acdnt_other_than_hit_and_run_  \
count                     5322.00      5322.00                        5322.00
mean                         4.03        50.88                         102.23
std                          7.30        95.28                         136.98
min                          0.00         0.00                           0.00
25%                          0.00         0.00                           0.00
50%                          1.00         7.00                          45.00
75%                          5.00        54.00                         165.00
max                        184.00      1089.00                        1553.00


Columns:
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hit_and_run_', 'deaths_negl_rel_rail_acdnt', 'deaths_due_med_negnc', 'deaths_neg_civic_bodies', 'deaths_other_negnc', 'dowry_deaths', 'abetment_of_suicide', 'atmpt_cmmt_murde
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rt_endgrng_lf_safety', 'other_simple_hurt', 'vlntrly_caus_grvus_hurt', 'grvus_hrt_dangerous_wepn', 'acid_attack', 'atmpt_acid_attack', 'grvus_hrt_pub_srvnt_duty', 'grvus_hrt_l
f_safety', 'other_grievous_hurt', 'wrngful_rest_cnfm', 'assault_on_women', 'sex_hrrsmt_work_office_prms', 'sex_hrrsmt_pub_trnsprt_sys', 'sex_hrrsmt_shelter_homes', 'sex_hrrsmt
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arian', 'rioting_industrial', 'rioting_political', 'rioting_caste_conflict', 'rioting_agrarian', 'rioting_students', 'rioting_vigilants', 'rioting_money_dispute', 'rioting_wat
er_dispute', 'rioting_land_property_dispute', 'rioting_family_disputes', 'rioting_enmity_rivalry', 'riot_elec_pwr_disp', 'riot_andolan_morcha', 'riot_polc_prsnl_gvt_srvnt', 'o
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rvng_pblc_wy', 'grvus_hrt_rsh_nglgnt_drvng', 'obstruction_pub_way', 'sale_obscene_books_objects', 'obscene_acts_songs_public_places', 'ofnc_religion', 'cheating_impersonatio
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s']


Missing Values per Column:
rioting_vigilants    3614
atmpt_acid_attack       1
dtype: int64

Total Missing Values: 3615


Cleaned dataset shape: (5322, 123)
```
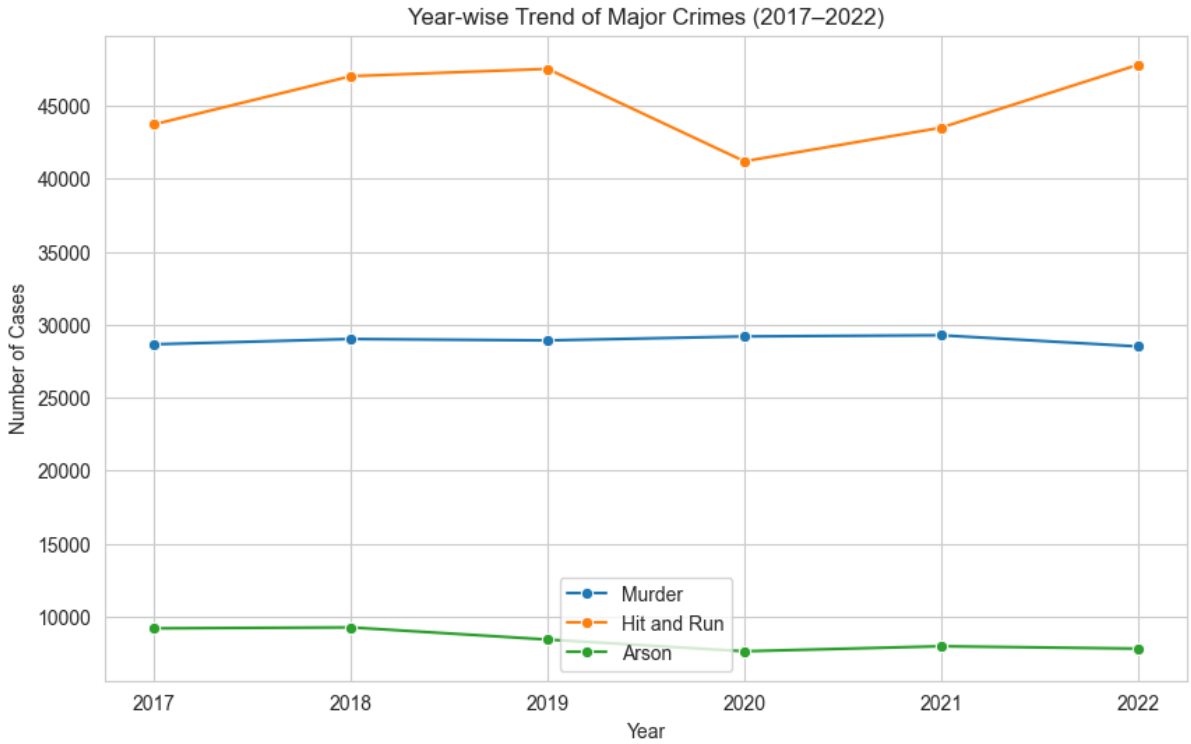
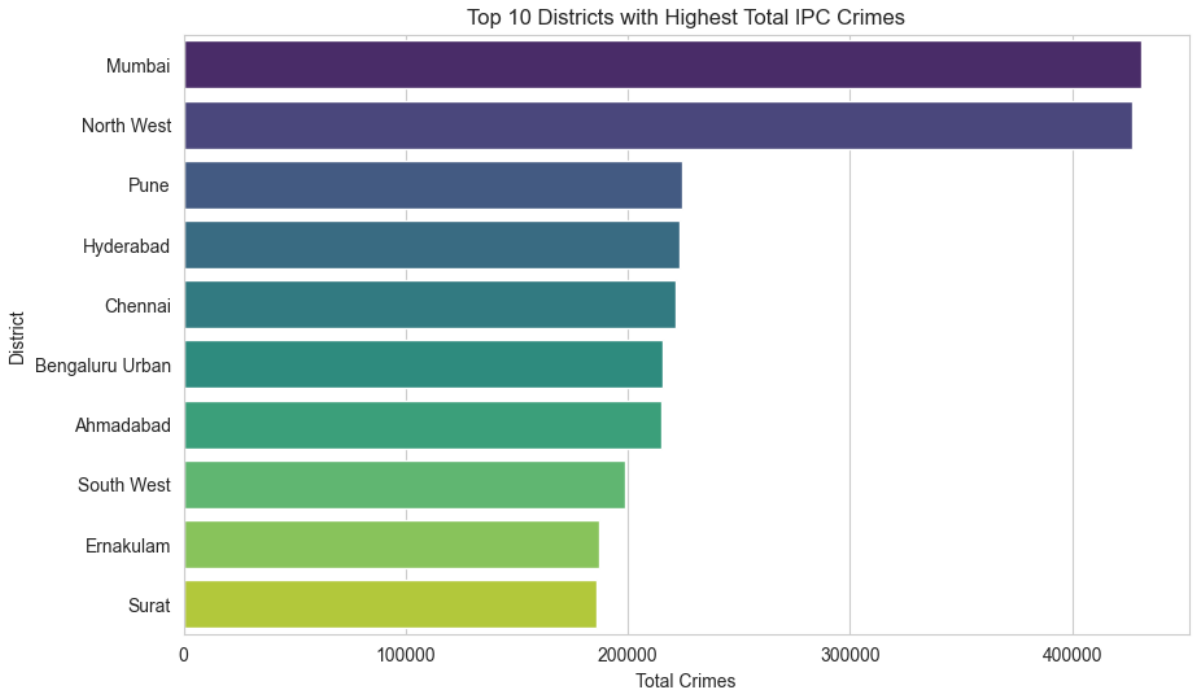
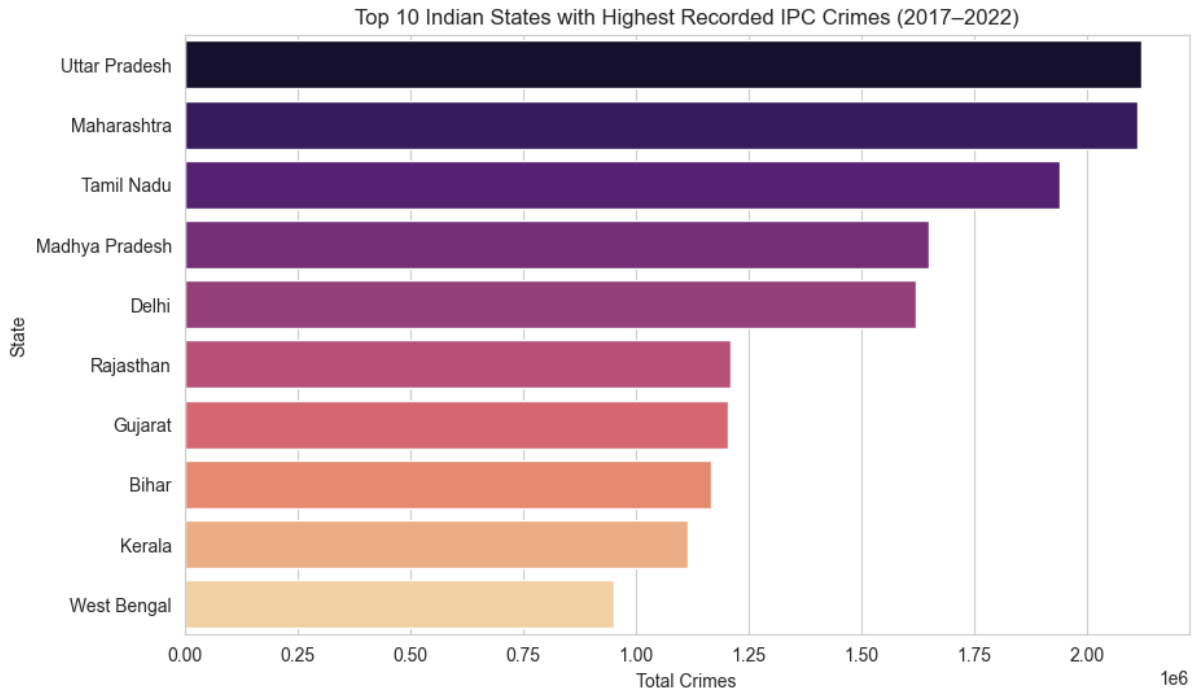
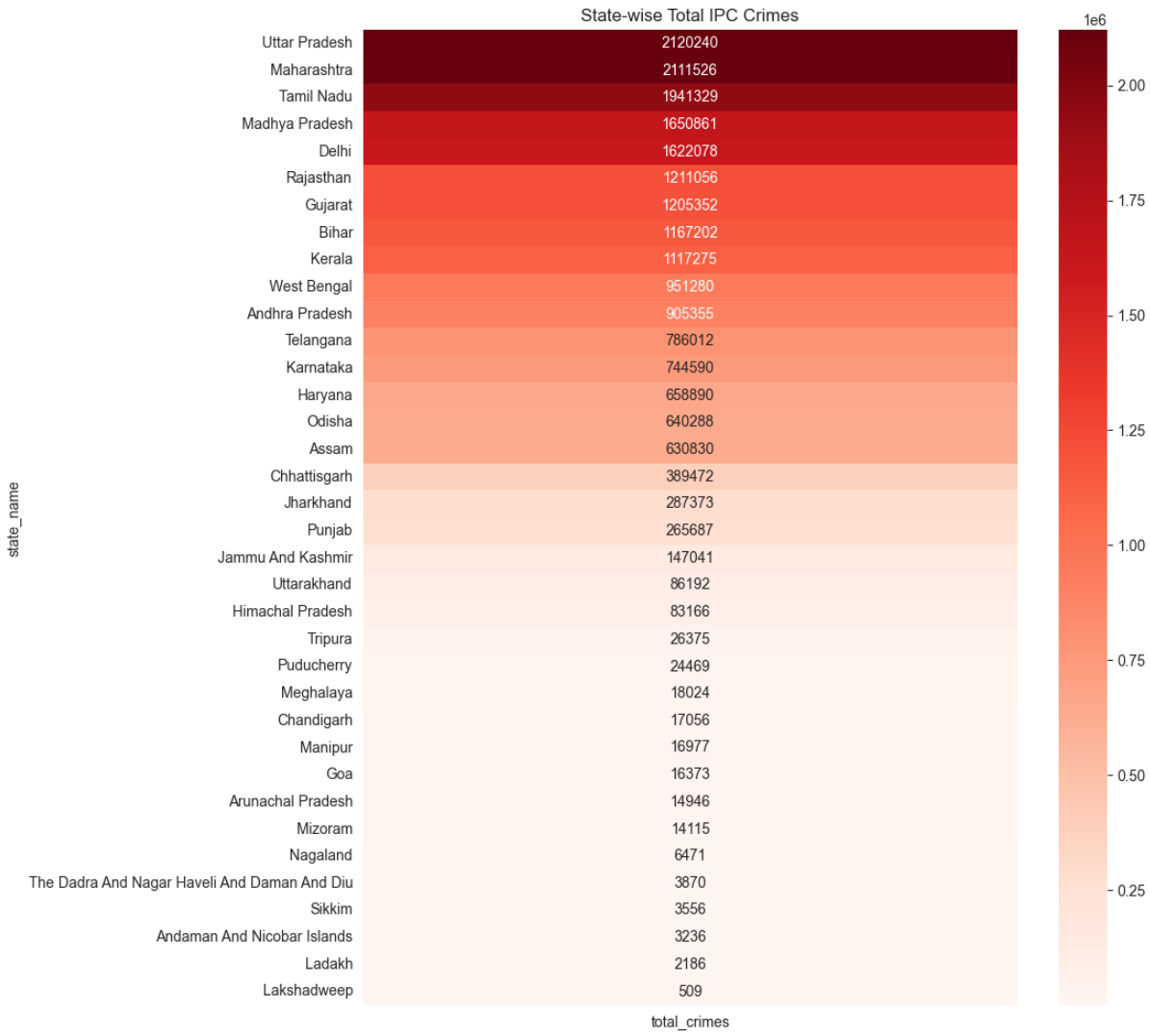
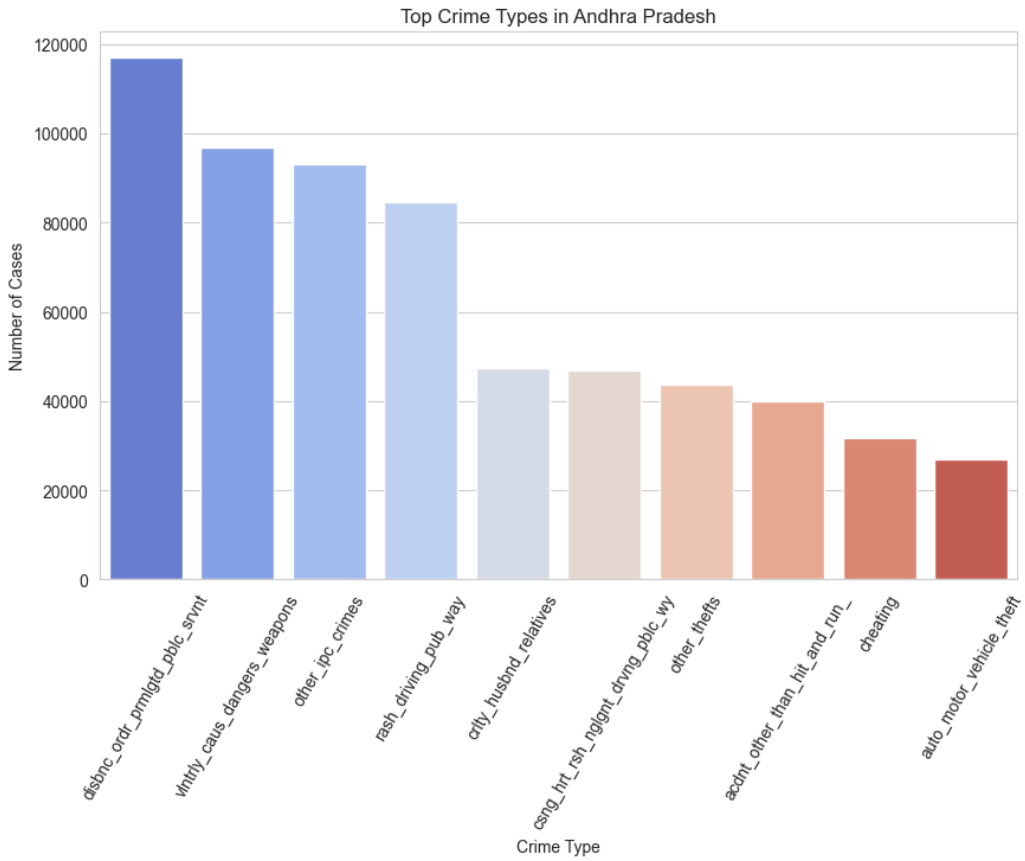

	state_name	district_name	year	total_crimes
0	Andhra Pradesh	Ananthapuramu	2017	8119.0
1	Andhra Pradesh	Chittoor	2017	3804.0
2	Andhra Pradesh	Y.S.R.	2017	14502.0
3	Andhra Pradesh	East Godavari	2017	9041.0
4	Andhra Pradesh	Ananthapuramu	2017	751.0

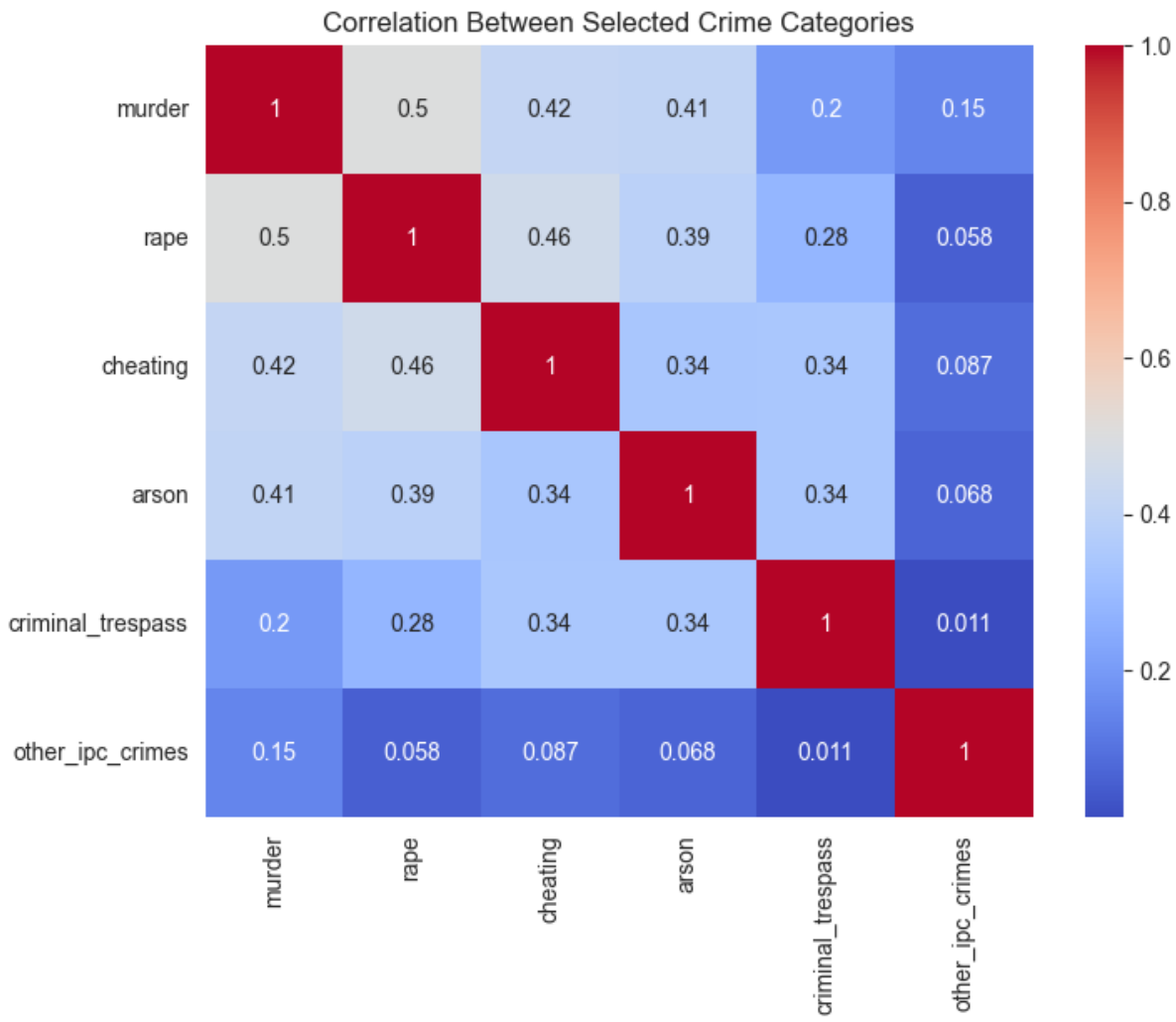

Year-wise Trend of Major Crimes (2017–2022)
Number of Cases
45000
40000
35000
30000
25000
20000
15000
10000
Murder
Hit and Run
Arson
2017
2018
2019
2020
2021
2022
Year

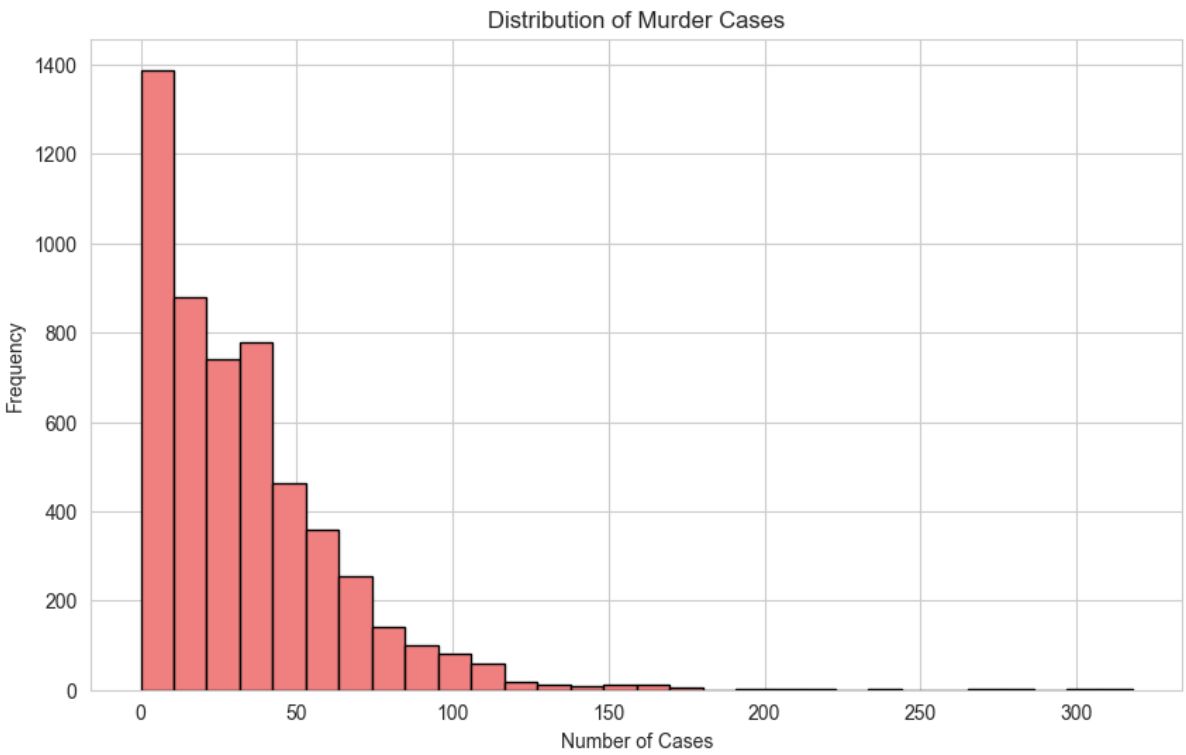

Top 10 Districts with Highest Total IPC Crimes
District
Mumbai
North West
Pune
Hyderabad
Chennai
Bengaluru Urban
Ahmadabad
South West
Ernakulam
Surat
0
100000
200000
300000
400000
Total Crimes

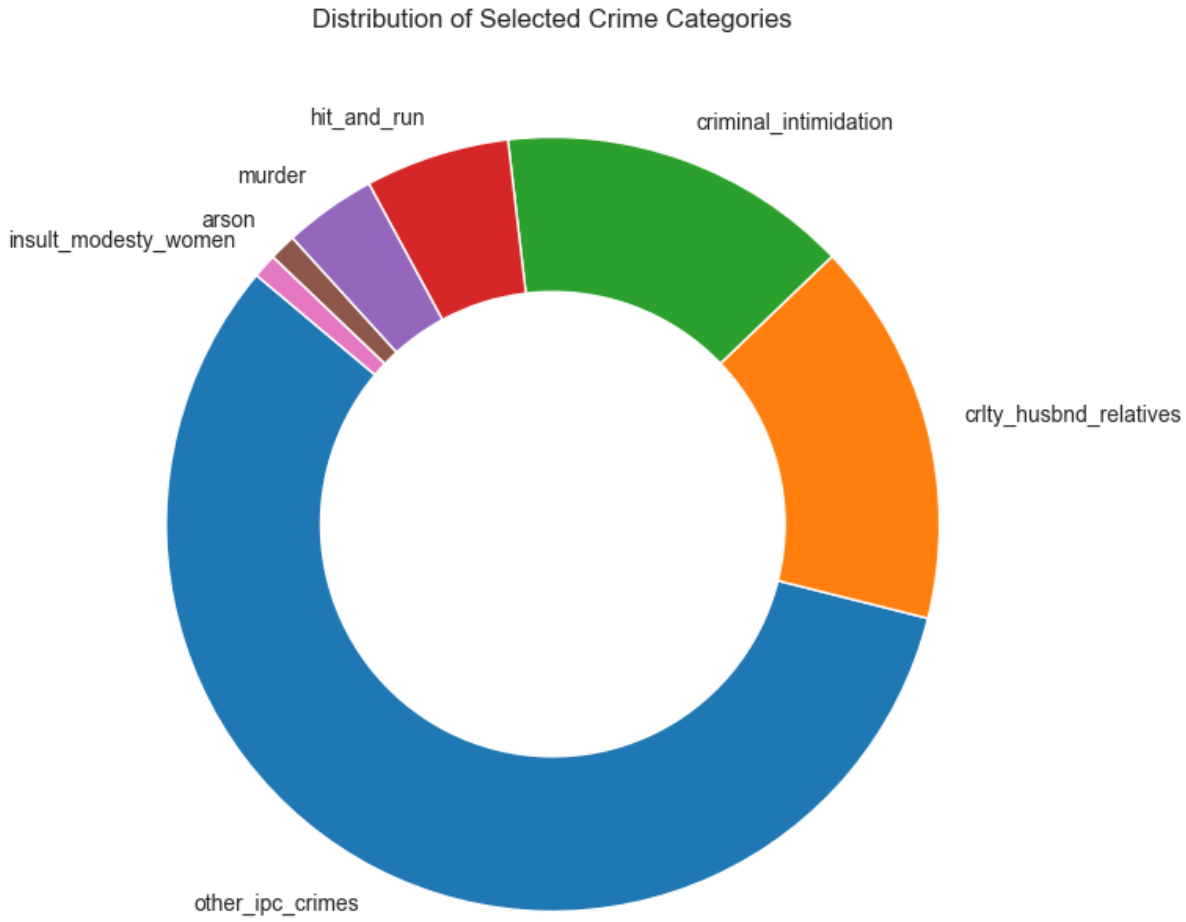

Top 10 Indian States with Highest Recorded IPC Crimes (2017–2022)
Uttar Pradesh
Maharashtra
Tamil Nadu
Madhya Pradesh
Delhi
Rajasthan
Gujarat
Bihar
Kerala
West Bengal
State
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Total Crimes
1e6


State-wise Total IPC Crimes
Uttar Pradesh 2120240
Maharashtra 2111526
Tamil Nadu 1941329
Madhya Pradesh 1650861
Delhi 1622078
Rajasthan 1211056
Gujarat 1205352
Bihar 1167202
Kerala 1117275
West Bengal 951280
Andhra Pradesh 905355
Telangana 786012
Karnataka 744590
Haryana 658890
Odisha 640288
Assam 630830
Chhattisgarh 389472
Jharkhand 287373
Punjab 265687
Jammu And Kashmir 147041
Uttarakhand 86192
Himachal Pradesh 83166
Tripura 26375
Puducherry 24469
Meghalaya 18024
Chandigarh 17056
Manipur 16977
Goa 16373
Arunachal Pradesh 14946
Mizoram 14115
Nagaland 6471
The Dadra And Nagar Haveli And Daman And Diu 3870
Sikkim 3556
Andaman And Nicobar Islands 3236
Ladakh 2186
Lakshadweep 509
state_name
total_crimes
1e6
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25


Top Crime Types in Andhra Pradesh
Number of Cases
120000
100000
80000
60000
40000
20000
0
dsbnc_ordr_prmlgtd_pblc_srvnt
vntrtly_caus_dangers_weapons
other_ipc_crimes
rash_driving_pub_way
crlty_husbnd_relatives
csng_hrt_rsh_nglgnt_drvng_pblc_wy
other_thefts
acdnt_other_than_hit_and_run_
cheating
auto_motor_vehicle_theft
Crime Type


Correlation Between Selected Crime Categories
murder 1 0.5 0.42 0.41 0.2 0.15
rape 0.5 1 0.46 0.39 0.28 0.058
cheating 0.42 0.46 1 0.34 0.34 0.087
arson 0.41 0.39 0.34 1 0.34 0.068
criminal_trespass 0.2 0.28 0.34 0.34 1 0.011
other_ipc_crimes 0.15 0.058 0.087 0.068 0.011 1
murder
rape
cheating
arson
criminal_trespass
other_ipc_crimes
1.0
0.8
0.6
0.4
0.2


Distribution of Murder Cases
Frequency
Number of Cases
0
200
400
600
800
1000
1200
1400
0
50
100
150
200
250
300


Distribution of Selected Crime Categories
hit_and_run
criminal_intimidation
murder
arson
insult_modesty_women
crlty_husbnd_relatives
other_ipc_crimes

7. CONCLUSION

This project presents a comprehensive exploratory data analysis of district-wise IPC crime data in India from 2017 to 2022. The objective of the study was to analyze crime trends, identify high-crime regions, examine relationships between different crime categories, and understand the distribution of major crimes across districts and states. Before conducting the analysis, several preprocessing steps such as handling missing values, removing redundant columns, eliminating duplicate records, and validating numerical values were performed to ensure the reliability and consistency of the dataset.

Various data visualization techniques—including line plots, bar charts, heatmaps, histograms, and pie charts—were used to reveal patterns and insights within the data. The year-wise trend analysis of major crimes such as murder, hit-and-run incidents, and arson indicated that these crimes remained relatively stable over the observed period, with minor fluctuations across different years. The analysis of district-level and state-level crime totals highlighted that large metropolitan districts such as Mumbai, Pune, Hyderabad, Chennai, and Bengaluru Urban recorded some of the highest crime counts. Similarly, states including Uttar Pradesh, Maharashtra, and Tamil Nadu exhibited the highest overall crime totals in the dataset.

The heatmap visualization demonstrated the uneven distribution of crimes across different states, emphasizing the concentration of crime incidents in densely populated regions and major urban areas. In addition, the distribution analysis showed that certain crime categories—particularly other IPC crimes, theft-related offences, and domestic cruelty cases—constitute a significant portion of the overall crime statistics. The correlation analysis further indicated moderate relationships among some crime categories, suggesting that certain offenses may share similar social or situational factors.

Overall, this project demonstrates the importance of data-driven approaches in understanding crime patterns and regional crime dynamics. The insights derived from this analysis can support law enforcement agencies, policymakers, and researchers in developing targeted crime prevention strategies, improving resource allocation, and enhancing public safety initiatives. Furthermore, this exploratory analysis provides a foundation for future work, including predictive crime modelling and advanced criminological studies using machine learning techniques.

8. FUTURE SCOPE

This exploratory analysis of district-wise IPC crime data provides a strong foundation for future developments in both criminological research and data-driven policy planning. While the current study focuses on identifying crime trends and patterns using exploratory data analysis techniques, several advanced analytical directions can be pursued to enhance the scope and impact of the research.

One important future direction is the application of machine learning and predictive analytics to forecast crime-prone regions and detect emerging crime patterns. Predictive models such as regression models, decision trees, and clustering techniques could be used to anticipate potential crime hotspots and assist law enforcement agencies in proactive planning and resource allocation.

Another promising extension involves the use of geospatial analysis and Geographic Information Systems (GIS) to visualize crime distribution across geographical locations. Mapping crime data spatially can help reveal regional patterns, urban–rural disparities, and localized crime concentrations. Such visualizations would enable administrators and policymakers to identify vulnerable areas and deploy security measures more effectively.

Future research could also incorporate socioeconomic and demographic variables such as population density, unemployment rates, education levels, and economic conditions. Combining crime data with these external factors may provide deeper insights into the underlying causes of criminal activity and help develop more targeted crime prevention strategies.

In addition, the development of interactive data dashboards using platforms such as Tableau, Power BI, or Streamlit could significantly improve the accessibility and usability of crime data. Interactive dashboards would allow policymakers, researchers, and the general public to explore crime patterns dynamically and gain real-time insights into regional crime trends.

Overall, extending this project with advanced analytical methods, geospatial visualization, and interactive reporting tools could greatly enhance the effectiveness of data-driven crime analysis and support more informed decision-making in the field of public safety and crime prevention.

9. REFERENCES

[1] **National Crime Records Bureau (NCRB), Government of India.**
Official Crime Statistics and Reports.
Available at: https://ncrb.gov.in
(Primary source for national crime statistics, crime classifications, and historical crime records in India.)

[2] **Open Government Data (OGD) Platform India.**
District-wise IPC Crime Dataset.
Available at: https://data.gov.in
(Official repository providing publicly accessible datasets, including district-level IPC crime statistics used in this study.)

[3] **Ministry of Home Affairs, Government of India.**
Crime in India Annual Reports.
Available at: https://mha.gov.in
(Official reports and publications that provide insights into crime trends, policy frameworks, and law enforcement statistics in India.)

10. LINKEDIN AND GITHUB

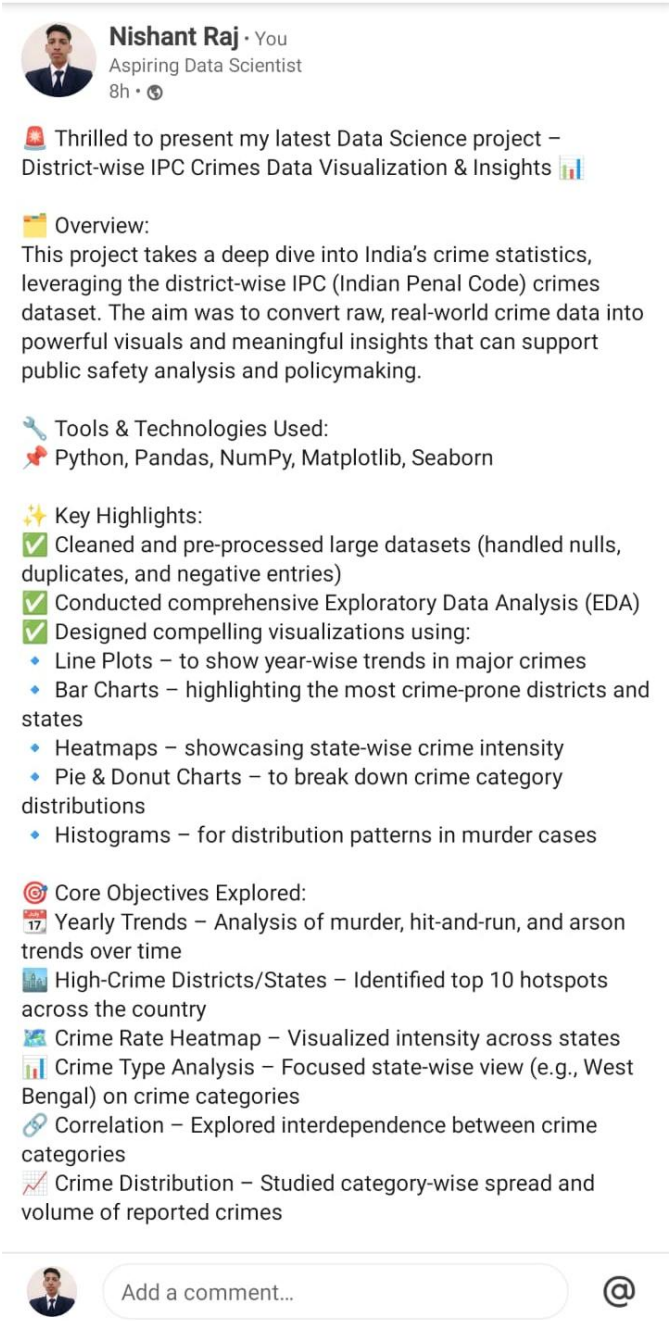
LinkedIn Link:

https://www.linkedin.com/posts/nishantrajx_datascience-python-eda-activity-7316732130980478976-5CIy?utm_source=share&utm_medium=member_android&rcm=ACoAAEbk7sQBXGzMPWRpwFFMPMFzWdqTb7h3LOg

GitHub Link:

https://github.com/nishantrajx/IPC-crime-data-analysis

LinkedIn Photos



Nishant Raj • You
Aspiring Data Scientist
8h •

🚨 Thrilled to present my latest Data Science project – District-wise IPC Crimes Data Visualization & Insights 📊

🗂️ Overview:
This project takes a deep dive into India's crime statistics, leveraging the district-wise IPC (Indian Penal Code) crimes dataset. The aim was to convert raw, real-world crime data into powerful visuals and meaningful insights that can support public safety analysis and policymaking.

🔧 Tools & Technologies Used:
📌 Python, Pandas, NumPy, Matplotlib, Seaborn

✨ Key Highlights:
✅ Cleaned and pre-processed large datasets (handled nulls, duplicates, and negative entries)
✅ Conducted comprehensive Exploratory Data Analysis (EDA)
✅ Designed compelling visualizations using:
🔹 Line Plots – to show year-wise trends in major crimes
🔹 Bar Charts – highlighting the most crime-prone districts and states
🔹 Heatmaps – showcasing state-wise crime intensity
🔹 Pie & Donut Charts – to break down crime category distributions
🔹 Histograms – for distribution patterns in murder cases

🎯 Core Objectives Explored:
📅 Yearly Trends – Analysis of murder, hit-and-run, and arson trends over time
🏙️ High-Crime Districts/States – Identified top 10 hotspots across the country
🗺️ Crime Rate Heatmap – Visualized intensity across states
📊 Crime Type Analysis – Focused state-wise view (e.g., West Bengal) on crime categories
🔗 Correlation – Explored interdependence between crime categories
📈 Crime Distribution – Studied category-wise spread and volume of reported crimes

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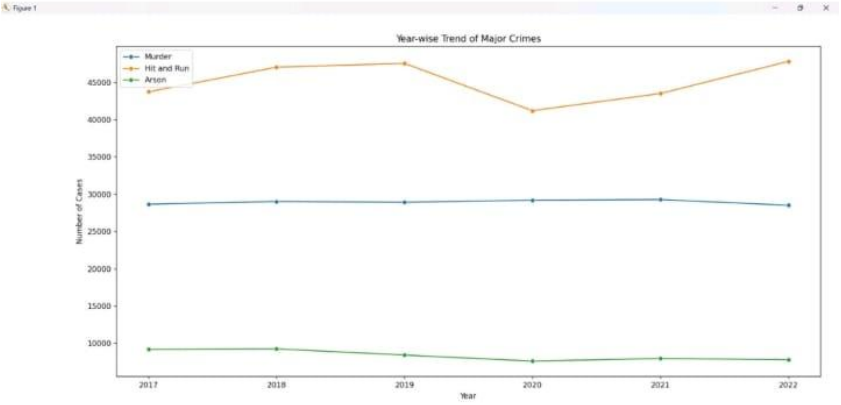
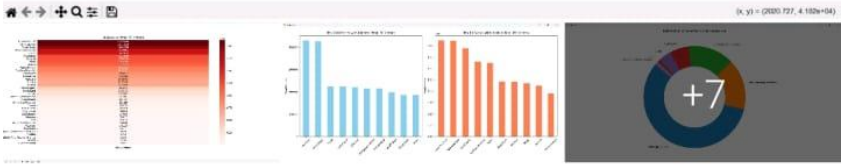
23:36 1 device

#DataScience #Python #EDA #Visualization #IPC #CriminalData #SocialImpact #Governance #Criminology #PublicPolicy #DataForChange #LinkedInPost #LPU #BtechCSE #SCSE





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